Analysis of Reading Skill and Contributing Factors

Spring 2026

Abstract

This research project sought to identify which factors have a significant impact on reading outcomes such as comprehension and reading speed. As a result we hoped to determine how one might best be able to improve their reading ability. We specifically analyzed the impact that the format of the text and the presence of music had on our reading outcomes. To address these questions we formulated a balanced, incomplete, factorial, Latin square design. We then fit both mixed and fixed effects models who were examined using analysis of variance methods. We can concluded that there was insufficient evidence to detect a significant effect on reading comprehension for any of our factors. However, we found that music had a significant effect on the total reading time. These results were somewhat consistent with our hypotheses and we believe that this experiment and design when scaled larger could produce even more meaningful results.

Introduction

Reading is something that most people do every single day whether it is for enjoyment or an assigned reading for work or school. We were interested in finding how someone might best be able to improve their reading ability (measured by both speed and comprehension). We are also interested if some factors have a greater impact on one of these impacts than on the other. We decided to study the format in which a text is read (either printed on paper or digitally on a laptop) and whether the participant was listening to music as they read. Nowadays digital work is becoming work common and it would be interesting to examine the impact that this has on reading ability. We also decided to include music as a factor because many people choose to listen to music as they work and our results may be generalizable to any sort of background music. In order to analyze our data, we fit mixed effects models and used analysis of variance procedures these were selected as a result of the experimental design and also because we wanted to include random effects in our models. we studied two separate response variables, fitting models for each of them because we believed them to be equally important and impossible to study separately. These response variables were the total reading time in seconds as well as the comprehension of the text (measured by the percentage score obtained on a four question quiz after reading). We predict that faster reading times may be associated with lower comprehension, which also influenced our decision to study both responses.

Experimental Design

In order to analyse our main research questions we formulated an experiment with a balanced, incomplete, factorial, Latin square design. We recruited twelve participants who each read three selected texts in a random order with random treatment levels selected. We only had three texts of similar length of difficulty available for this study. Due to this limitation and because seeing a text more than once would directly impact reading speed and comprehension we needed to use an incomplete design. We studied two factors with two levels each and blocked based on both the text read as well as the participant who was reading. Recruiting tweleve participants allowed us to obtain a greater sample size and record three repeated measurements which increases the sample size and improves the power of our tests.

Statistical Analyses

Model 1: Time

First we fitted a full factorial linear model including format, music, text, and participant as fixed effect to explore the overall patterns in the data. An ANOVA was created and provided a significant main effect of music (p=0.04867), text (p=0.02744), and participants (p=0.01437). This is something we expected before analyzing. Also, we saw that no interaction effects were significant.

However, because participants and text represent a random sample, we then fitted a mixed-effects model treating participants and text as our random effects. We also wanted to explore participant specific random slope for music. Changing our model to this allows for results to be generalized beyond our specific data collection.

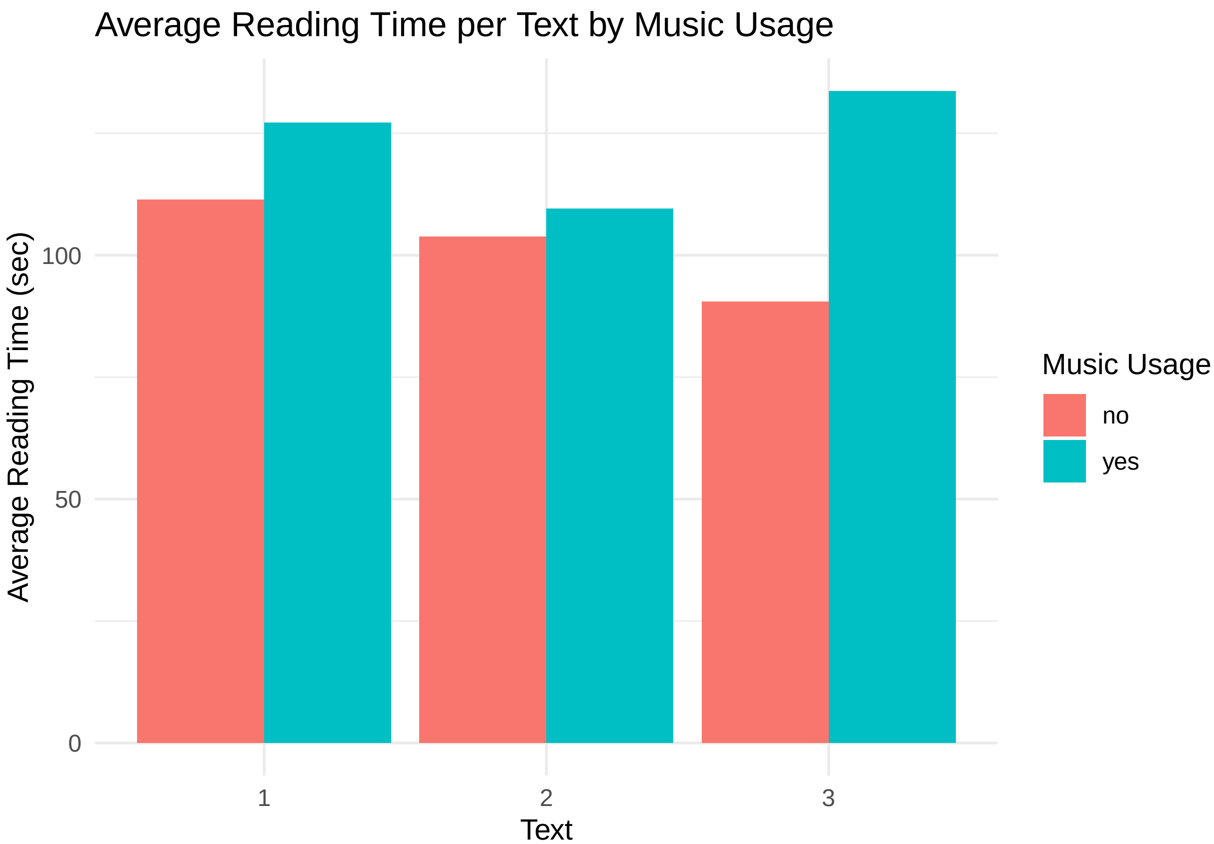
Write model here:

From the ANOVA of this model, we concluded that music was the only statistically significant main effect with a p-value of 0.0112. There is no significant effect of the interaction term between format and music. Also, looking at the RANOVA p-values for our random effect factors we concluded that the random effect for text showed no evidence of variance between the 3 text. Additionally, the random effect of music by participant showed no evidence of variance between how music effects the reader. This shows that participants are fairly consistent and text does not explain much variance.

Table 1: Total Reading Time P-Values

Factor	P.Value
format	0.4117
music	0.0112
format:music	0.8768
(1 | Text)	1.0000
(1 + music | Participant)	0.7282

We want to visually show what the effect of music on total reading time looks like. Displayed below is a bar plot showing the effect of music on reading time per text. We see that for all 3 text, reading time increased with music usage.



Model 2: Comprehension

First we fitted a full factorial linear model including format, music, text, and participant as fixed effect to explore the overall patterns in the data. An ANOVA was created and provided no significant effect on comprehension for any main effect. We believe this is due to the lack of variance in our comprehensions scores.

Again, because participants and text represent a random sample, we then fitted a mixed-effects model treating participants and text as our random effects.

Write model here:

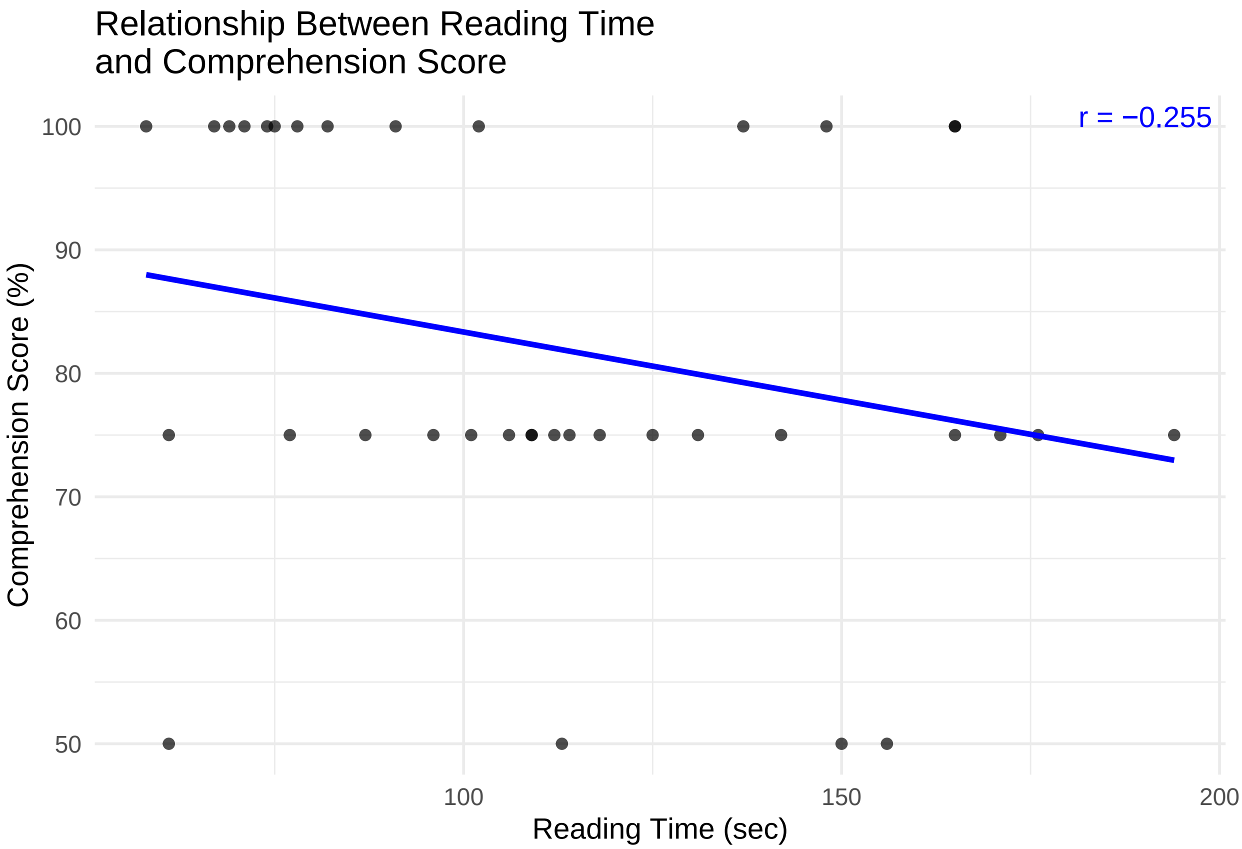
From this we concluded only the interaction between format and music showed statistical significant effect on comprehension with a p-value of 0.03796. No main effect showed a significant effect which is evidence for us to believe there is some lack of variance in our comprehensions scores. Additionally, consistent with the time model, the random effects did not provide significant evidence of explaining additional variance.

Table 2: Total Reading Time P-Values

Factor	P.Value
format	0.60384
music	0.94230
format:music	0.03796
(1 | Text)	1.00000
(1 + music | Participant)	0.92720

Since the comprehension model does not lead to very interesting analysis of results, we have decided to fit another model treating it as a fixed effect in order to determine if it had any effect on reading time. We would expect that lower comprehension score would be correlated with lower reading time. We must acknowledge however, that this is not a causal relationship but can be indicative of a relationship between these variables.

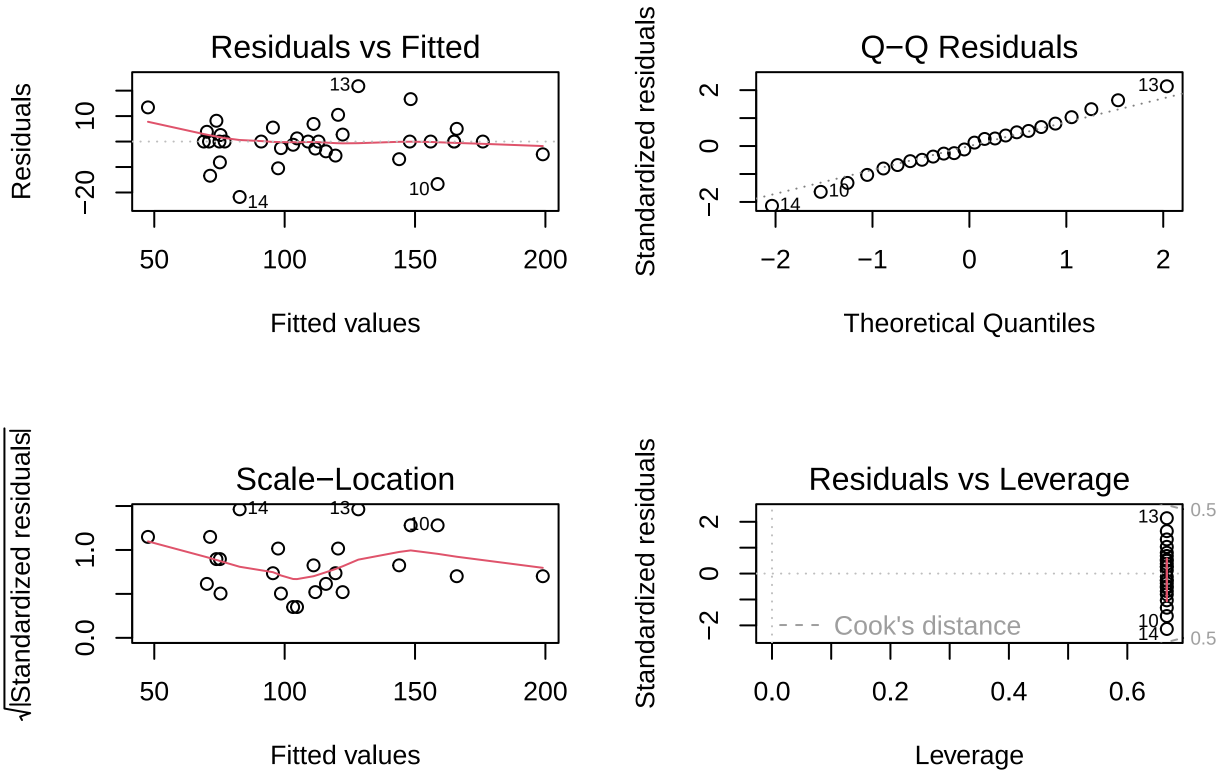
From this model again we concluded no main effect was statistically significant. Although comprehension is not a statistically significant predictor, the scatter plot with a regression line indicates a negative association between comprehension score and reading time.




```r
par(mfrow = c(2,2))
plot(model_time)
```

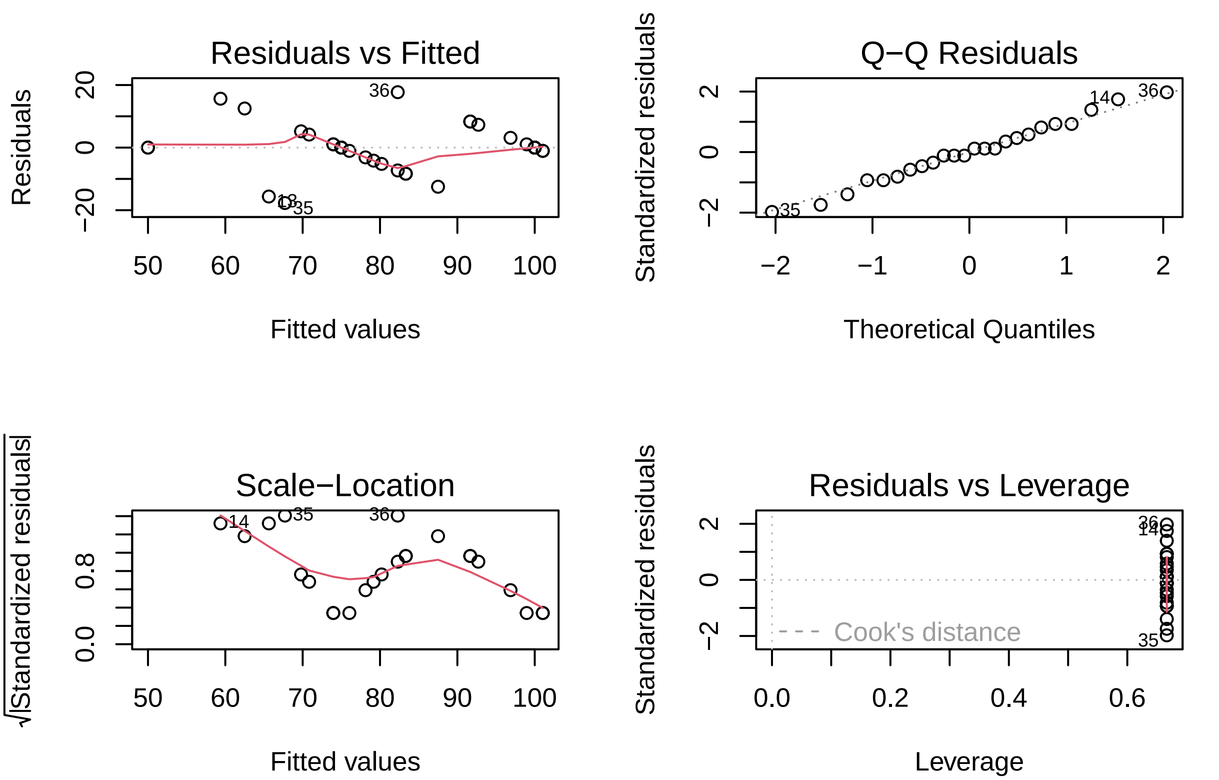
Diagnostics for ANOVA

```
## Warning: not plotting observations with leverage one:
##   2, 5, 8, 11, 15, 18, 21, 24, 25, 28, 31, 34
```




```r
par(mfrow = c(2,2))
plot(model_comp)
```

```
## Warning: not plotting observations with leverage one:
##   2, 5, 8, 11, 15, 18, 21, 24, 25, 28, 31, 34
```



Diagnostics for mixed-effects models

```r
isSingular(model_time_lmer)
```

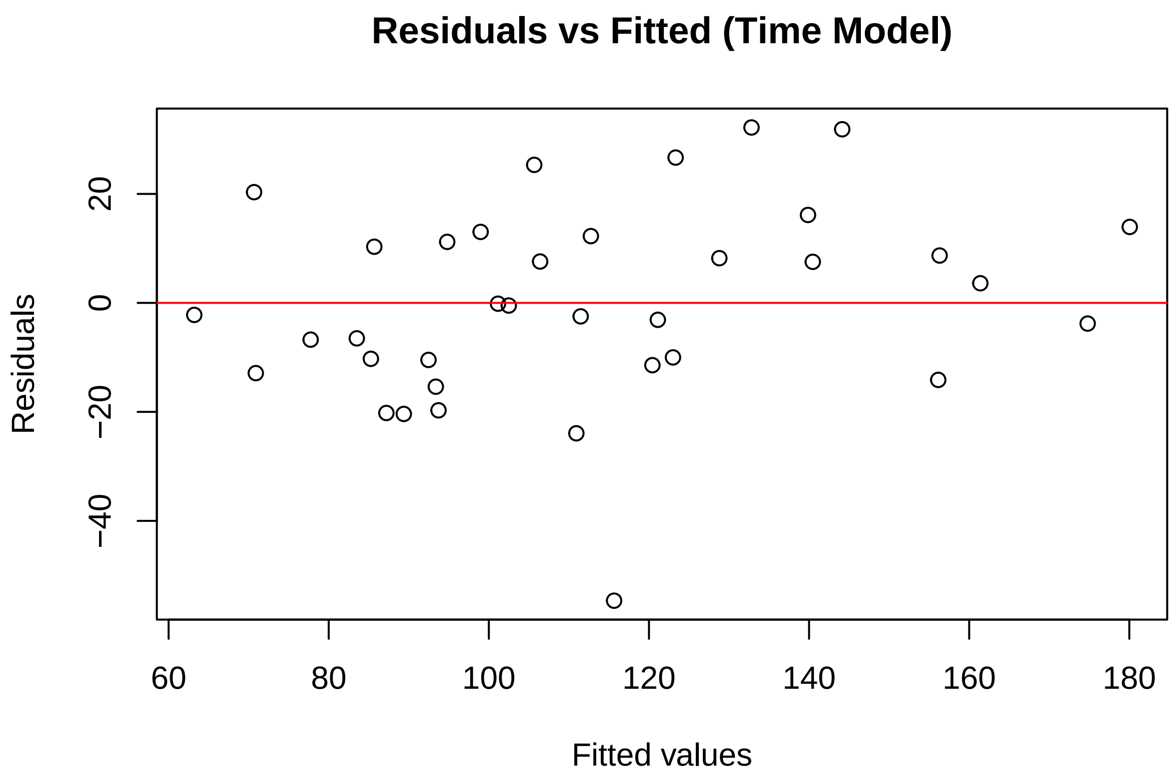
```
## [1] TRUE
```

```r
isSingular(model_comp_lmer)
```

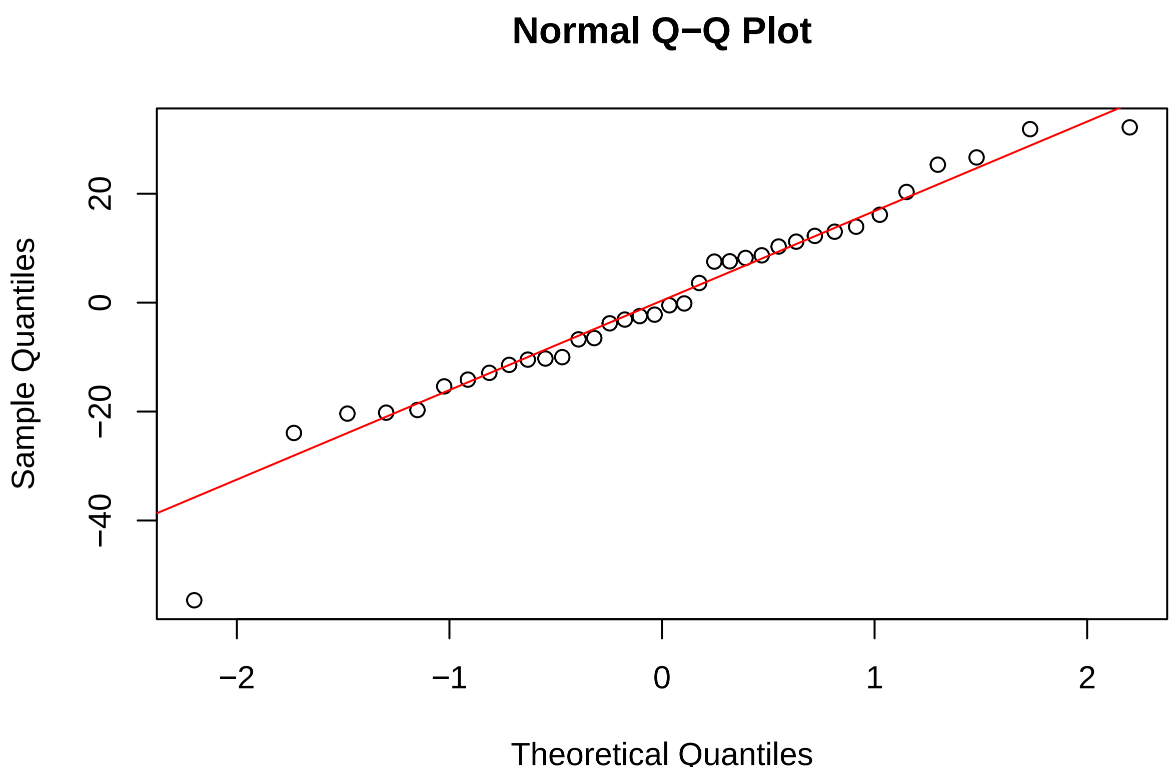
```
## [1] TRUE
```

```r
# Residual diagnostics: reading time mixed model

plot(fitted(model_time_lmer), resid(model_time_lmer),
     xlab="Fitted values",
     ylab="Residuals",
     main="Residuals vs Fitted (Time Model)")
abline(h=0,col="red")
```

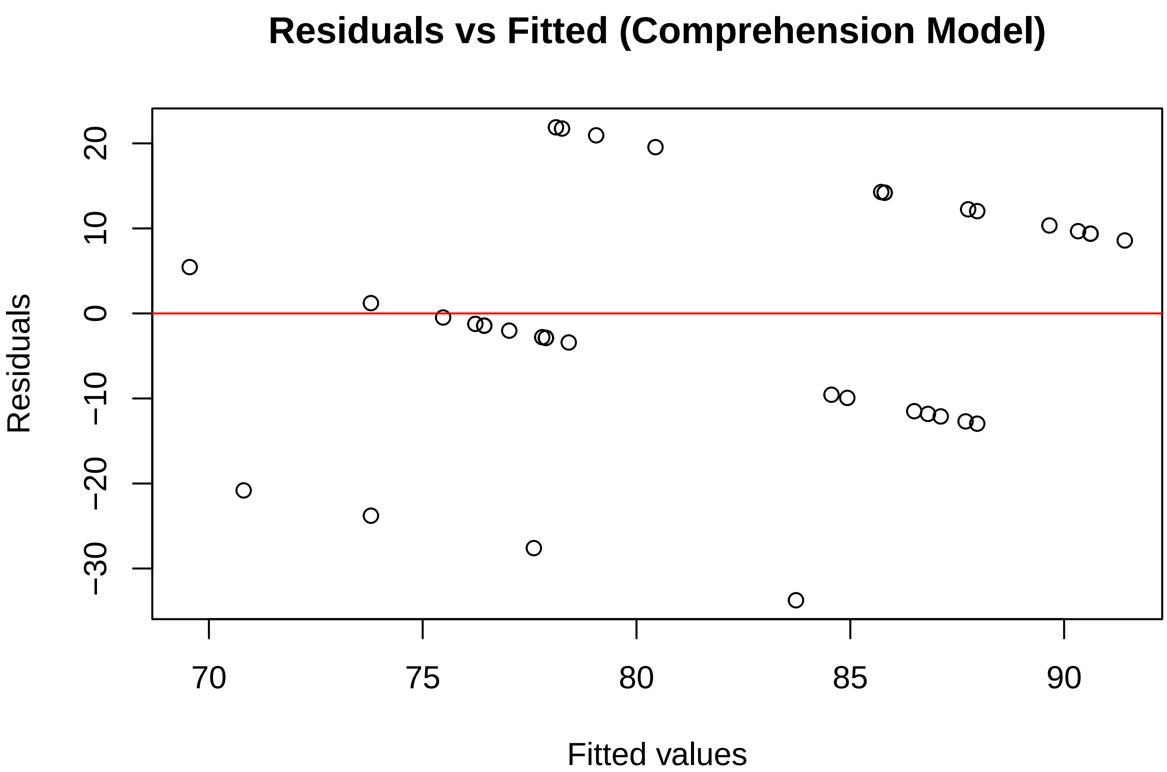



```r
qqnorm(resid(model_time_lmer))
qqline(resid(model_time_lmer), col="red")
```

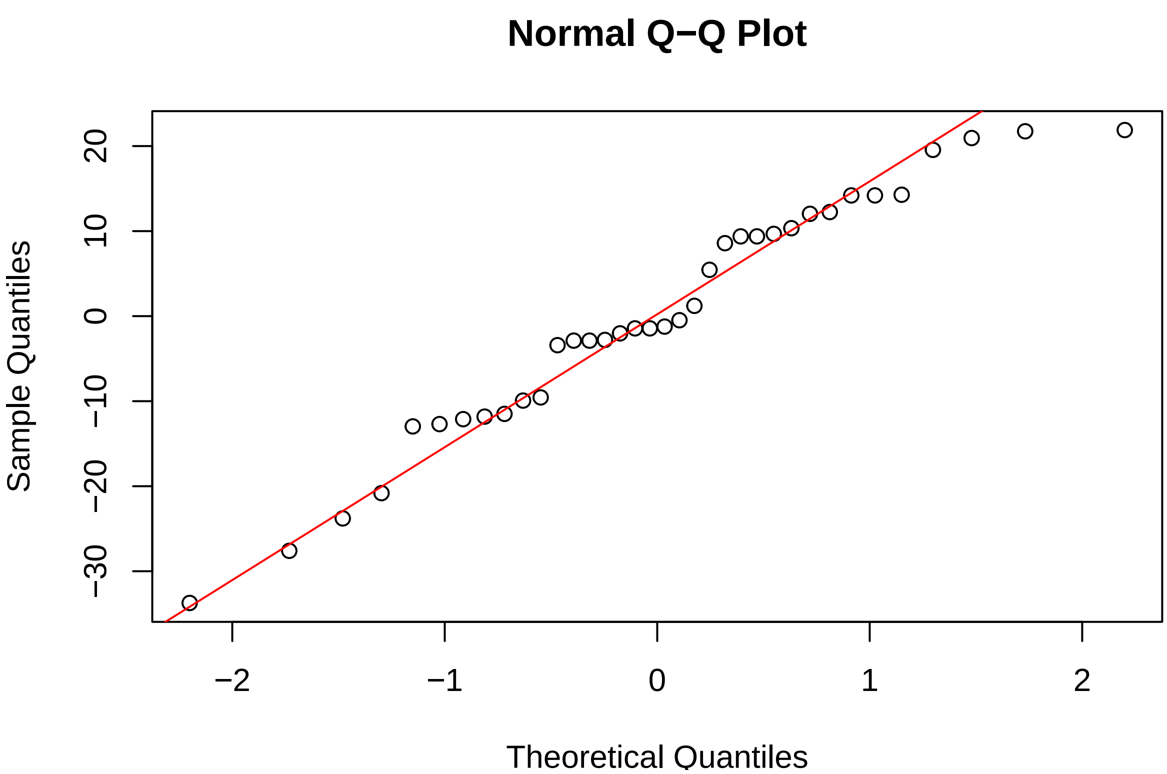



```
# Residual diagnostics: comprehension mixed model

plot(fitted(model_comp_lmer), resid(model_comp_lmer),
     xlab="Fitted values",
     ylab="Residuals",
     main="Residuals vs Fitted (Comprehension Model)")
abline(h=0,col="red")
```




```r
qqnorm(resid(model_comp_lmer))
qqline(resid(model_comp_lmer), col="red")
```



The diagnostic plots were used to assess model assumptions.

For the reading time model, the residuals versus fitted plot does not show a strong systematic pattern, suggesting that the assumption of constant variance is reasonable. The Q-Q plot indicates approximate normality of residuals with only minor deviations in the tails.

For the comprehension score model, the residual plots show slightly more limitation because the response variable takes only a small number of discrete values. However, no strong violation of modeling assumptions is observed.

The mixed-effects models produced singular fit warnings, indicating that at least one random-effect variance component is estimated close to zero. This suggests that the participant-specific slope for music contributes little additional variability beyond the intercept-only random effect. However, the mixed-effects structure was retained to reflect the experimental design.

Overall, diagnostics suggest that the reading time model is appropriate for inference, while the comprehension model should be interpreted more cautiously due to limited response variability.

Conclusions

From this analysis, we can conclude that there was insufficient evidence to detect a significant effect on reading comprehension. However, we found that music had a significant effect on total reading time. Thus, if you need to complete a reading in a specific amount of time, we have reason to believe that you shouldn't be listening to music. Though, our results were not too groundbreaking, we believe that our model is a good starting point for future exploration. With more varied data, more time to collect responses, and possible adjustments to the complexity of the model it would be interesting to investigate whether additional factors influence these responses or whether stronger effects emerge. This experiment largely focused on short passages but longer texts would be interesting to study and examine if our results hold. We believe that the format of the text would have a larger effect if the length of the text was increased because participants are more likely to experience screen fatigue or be otherwise distracted. Overall, this project provided a valuable exercise in applying mixed-effects modeling to real behavioral data.